

Que and Ans::

- **What is Electrical Engineering?**

Electrical Engineering is the field of Engineering that generally deals with the study and application of electricity, electronics, and electromagnetism.

Electrical engineering deals with high voltage especially AC above the range of 110 or 220V. Moreover, electrical components and devices use alternating current (AC) / Voltages and tend to be larger in size and require 230V (In UK) and 110V (In US) single-phase ac voltages whereas in industries and power stations, it may be up to 11kV and for transmission, it may be higher than 400kV

- 2. **Why Transformer Rated In kVA, Not in KW?**

When manufacturers design a transformer, they have no idea which kind of load will be connected to the transformer. The load may be resistive (R), inductive (L), capacitive (C) or mixed load (R, L, and C). Its mean, there would be different **power factor (p.f)** at the secondary (load) side. The output of real power may vary depending on the power factor. Thus the manufacturer denotes it as “this transformer can provide x number of amperes at y amount of voltage”. **This way, they go for VA as in (voltage x Amperes) instead of W in case of rating of a Transformer.**

- **3.What is an Ideal Transformer**

An **ideal transformer** is an imaginary transformer in which no losses occur at all. In other words, the **transformer** Input power is equal to the output power of the transformer i.e. they have 100% efficiency. It is just a theoretical transformer because there must be some **losses in a real transformer**.

Transformer input Power = Transformer Output Power.

$$P_{IN} = P_{OUT.}$$

- 4. **What are transformer losses & their types?**

A real transformer has the following losses;

- **Copper Losses (Winding Resistance)**

The losses occur due to the current flow through the windings that causes resistive heating of the conductors.

- **Core or Iron Losses**

There are two types of core or iron losses

- - **Hysteresis Losses**

These losses occur due to the magnetic field reversal. The magnetization & demagnetization of the core consumes some energy, also known as hysteresis loss.

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- **Eddy Current Losses**

The losses due to the current generated inside the core. The eddy current causes the resistive heating of the core.

Various other losses include:

- **Stray losses (leakage Flux)**
- **Dielectric Loss**
- **Other Losses**
 - **Magnetostriction Losses**
 - **Mechanical losses**
- **What is transformer efficiency & all-day efficiency? What is the condition for maximum efficiency?**

Transformer Efficiency:

The [efficiency of the transformer](#) is given by the output power divided by the input power. Some of the input power is wasted in internal losses of the transformer.

$$\text{Efficiency, } \eta = \text{Output Power} / \text{Input Power}$$

All Day Efficiency:

The ratio of energy delivered in Kilo Watt-Hour (kWh) to the energy input in kWh of the transformer for **24 hours** is called all-day efficiency.

$$\eta_{\text{all_day}} = \text{Output in kWh} / \text{Input in kWh}$$

Condition for Maximum Efficiency:

The copper loss must be equal to the iron loss; which is the combination of hysteresis loss & eddy current loss.

$$\text{Cu Loss} = \text{Iron Loss}$$

$$W_{\text{cu}} = W_{\text{i}}$$

- **What is the voltage regulation of the transformer & why is it important?**

The [voltage regulation of a transformer](#) is the percentage change in the secondary voltage from no load to full load condition. Ideally, the secondary voltage remains the same throughout the load, in which case the voltage regulation is zero. But practically it varies with the power factor of the load.

The voltage regulation value provides the efficiency of the transformer & it is best to prefer a transformer with low voltage regulation.

- **Why the current transformer's secondary should not be open when there is current flow in its primary?**

The current transformer is essentially a step-up transformer that increases the voltage & decreases the current on the secondary side. Under the open secondary condition, the primary current becomes the magnetizing current that generates a very high secondary voltage that can damage the insulation as well as can pose danger to personnel.

- **How do you know if a circuit is inductive or capacitive or purely resistive?**

The total impedance of the circuit can be used to identify the circuit. If the imaginary part of the total impedance is positive, then it is an inductive circuit. If the imaginary part is negative, and then it is a capacitive circuit. If it is zero then the circuit is purely resistive.

- **Does current lead or lag the voltage in an inductive or capacitive circuit?**

The current lags the voltage in an [inductive circuit](#) while the current leads the voltage in a [capacitive circuit](#).

- **What is a unilateral & bilateral circuit?**

A [unilateral](#) is a type of circuit whose properties change with the direction of current flow or the voltage. The properties of a bilateral circuit do not change with changing the current direction or supply voltage.

- **What is a linear & non-linear circuit?**

In a [linear circuit](#), the relation between the current & voltage is linear i.e. directly proportional. The circuit parameter such as frequency, resistance, inductance, capacitance, etc. remains constant with varying current & voltage.

While in the non-linear circuit, the current & voltage does not have a linear relationship. The electrical parameters of such circuits changes with varying voltage & current.

- **What are the limitations of ohm's law?**

[Ohm law](#) is not applicable to a unilateral circuit or a non-linear circuit. The criterion for ohm's law is that the resistance must be constant which also depends on the temperature. Whereas, the resistance of a non-linear or unilateral circuit varies with voltage & current. Thus it is not applicable in such a circuit. Also, the temperature must remain constant.

What is the maximum power transfer theorem?

It mentions the condition for [maximum power transfer](#) from source to load. It states that in a linear, bilateral network, the maximum power will be transferred from source to the

load when the external load resistance equals the internal resistance of the source or Thevenin's resistance of the circuit.

- **What is active, reactive, apparent & complex power?**

Active power: it is the actual power that is delivered to the load such as [transformer](#), induction motors, generators, etc and dissipated in the circuit. It is denoted by P & its unit is watts W.

Reactive Power: The powers that continuously bounce back and forth between source and load is known as reactive power. It is denoted by Q & its unit is VAR voltage-ampere reactive.

Apparent Power: It is the product of voltage & current without referencing the phase difference between them. It is a combination of active power & reactive power. It is denoted by S & its unit Volt-Ampere, VA.

Complex power: it is the product of voltage & current with reference to the phase difference between them. it is the complex sum or vector sum of the active power & reactive power. It is denoted by S & measure in VAR.

- **What is a leading & lagging power factor & how can you improve it? State the methods of power factor correction?**

The cosine of the angle between voltage & current is called the [power factor](#). If the current leads the voltage, it causes a leading power factor. If the current lags voltage the power factor is lagging.

Most loads are inductive that causes a lagging power factor. Thus these following methods are used for [p.f correction](#) (to nullify the lagging current)

- - **Static Capacitor**

The capacitor helps in providing a leading current that eliminates the lagging component of current & improves the power factor

- - **Synchronous Condenser**

It is an over-excited synchronous motor with no load that also provides a leading power factor.

- - **Phase Advancer**

Phase advancer is a simple AC exciter which is connected on the main shaft of the motor and operates with the motor's rotor circuit for power factor improvement. Phase advancer is used to improve the power factor of induction motor in industries.

- **Why do we improve the power factor?**

The reason for **improving the power factors** are stated below;

- **Large Line Losses (Copper Losses):** Line losses (I^2R) depend on current. The low power factor draws a large amount of current as compared to the high power factor.
- **Large kVA rating and Size of Electrical Equipment;** PF is inversely proportional to KVA. Low PF equipment with a high KVA rating is larger in size.
- **Large Conductor Size and Cost;** we need large conductors to transmit the heavy current required due to low power factor.
- **Poor Voltage Regulation and Large Voltage Drop;** The large current due to low PF causes a high voltage drop that needs to be regulated more often than usual.
- **Low Efficiency;** The losses due to the high current flow & voltage drop deteriorate the efficiency of the system. The efficiency is maximum at PF=1.

- **Why Motor rated in kW instead of kVA?**

As we know the transformer is rated in KVA because its PF (Power Factor) depends on the nature of the loads. However, **Motor** has a fixed **Power factor**, i.e. motor has defined power factor (P.F) and the rating has been mentioned in kW or HP on Motor. In more clear words, Motor only consumes active power and provides mechanical power in HP or kW at the motor shaft and that is the reason for **motor rating in Watts**.

- **What is a motor starter?**

Motor Starter is a device that connects in series with the motor to decrease the starting current (that could damage the windings in normal conditions) and gradually increase current after starting the motor (in other words start or stop the motor) and provide overload protection.

- **What is a Magnetic starter?**

A **magnetic starter** is a device designed to provide a safe starting method to electric motors with a heavy load. It includes a contactor as an essential component, while also providing power-cutoff, under-voltage, and overload protection.

- **Why is Battery Rating in Ah (Ampere hour) and not in VA or Watts?**

A battery converts chemical energy into electrical energy which is the charge stored inside the chemicals. The amount of current it can supply for a said time thus **Ampere-hour Ah is the unit for its rating**.

While the batteries supply direct current which has no phase or frequency thus there is no concept of P.F or reactive power, thus no need for expressing it in VA and its rated in Ah.

- **What is a primary & secondary cell?**

The primary cell is a non-rechargeable battery that cannot be recharged by any means. They are disposable & cannot be used once they are fully discharged. They are mostly used in toys, handheld devices & remote controllers, etc.

The secondary cell is a rechargeable battery that can be recharged several hundreds of times (depends on its life cycle). Their initial cost is expensive compared to the primary cell. They are mostly used in cellphones, vehicles, generators, etc. We have discussed both in details in [different types of batteries](#).

- **What is the difference between circuit breaker & Isolator?**

Circuit Breaker

A Circuit Breaker is a protective electromechanical device used to control the flow of current same like a [fuse](#). It automatically breaks the circuit in case of fault conditions like [short circuit](#) and overload. It can also manually break the circuit. It is ON-load & OFF load device, it means it operates in both ON/OFF supply condition.

Isolator:

An [isolator](#) is a mechanical switch used for isolating or disconnecting power supply in substations. It is an off-load device i.e. it is operated when the power supply is off.

- **What is the difference between MCB & MCCB?**

The MCB stands for “[miniature circuit breaker](#)” & it is used for current rating lower than 100 amps with interrupting ratings of below 18k Amps. Its tripping characteristics cannot be adjusted & they are used for domestic purposes.

The [MCCB](#) stands off “Molded case circuit breaker”. It has a high current rating of around 2500 Amps, where its interrupting ratings are between 10K to 200k Amps. Also, its tripping characteristics can be adjusted. They are used in industries.

- **What is the difference between a generator and an alternator?**

A generator & an alternator both convert the mechanical energy into electrical energy. The key difference between them is that the generator has a stationary magnetic field i.e. the armature rotates inside a fixed magnetic field. while in an alternator, the magnetic field rotates inside a conductor windings (stator). So the stationary part of the alternator is phase conductors.

- **Define the term Capacitance and Inductance**

Capacitance: capacitance is the ability of a component to store charge between two plates when there is a potential difference applied. It is denoted by C & it is measured in Farads F.

Inductance: inductance is the ability of a conductor to resist or oppose any change in the current. The current generates a magnetic field whose strength varies with the current. It is denoted by L & is measured in Henry H.

- **What are the different colors on wires indicates?**

The different colors of the wires are used for phase indication purposes. They represent different phases, the neutral & earth cable. The color code may differ around the world but usually, the earth wire remains the same i.e green with yellow stripes.

- **How do you decide what size of electrical wire do you need?**

The **proper size of the wire** mainly depends on the amount of load current & voltage supplied as well as the length of the cable. Usually, the voltage drop in cables should not be greater than 2.5% of the supply.

- Measure the supply voltage.
- Calculate its 2.5% allowable voltage drop.
- Measure the required length of the cable in meters.
- Divide the allowable voltage drop with the length.
- The resultant is the max allowable voltage-drop per meter & your required cable size should have a lower voltage drop than this.
- Calculate your load current.

The cable charts provide this info such as voltage-drop per meter, current capacity, cable size, etc. match your calculation with the chart & find you required cable size.

- **What are the types of semiconductors?**

The main two types of **semiconductors** are;

- **Intrinsic semiconductor;** the semiconductor that has same number of electrons & holes. They exist in their natural form.
- **Extrinsic semiconductor;** these semiconductors have foreign impurity (electrons or holes) inserted artificially through the process called doping. The impurities are intentionally added to change its electrical properties.
 - **N-Type:** The dopant in this type of semiconductor increases the number of free electrons. So the majority carriers are electrons in N-type.
 - **P-Type:** the dopant of such semiconductors adds in the excess number of holes. The majority carriers in P-type semiconductors in holes.

- **What is a transistor composed of?**

Transistor is made of different combinations of P-type & N-type semiconductors. The doping & combination & different shapes of semiconductors form different types of transistors with different electrical properties.

- **What is the role of a transistor in Circuit?**

Mainly the transistor can be used for two reasons.

- **Switching:** to switch ON/OFF the flow of current in a circuit. The switching depends on the input voltage or current.
- **Amplification:** to increase or amplify the input signal that has very low power into a high power signal.

- **What will be current if the Total resistance in a series circuit doubles?**

The current has a linear inverse relation with the resistance. Thus, doubling the resistance will decrease the current by half.

- **What could be the reason for the current to double in a linear circuit?**

There are two reasons for increasing the current;

- 1) Either the total resistance of the circuit is reduced by half
- 2) Or the supply voltage to the circuit is doubled.

- **What is meant by reverse polarity and how it can be fixed?**

An electrical outlet has two wires i.e. neutral & hot or live wire. The Reverse polarity means that the neutral wire is connected to the terminal where the hot wire is supposed to be. It can create shock hazards because the ON/OFF switch will cutoff only the neutral wire from the appliance connected to such outlets.

It can be easily fixed by swapping the wire.

- **Explain what rectifiers are and what are the types of rectifiers?**

A rectifier is an electronic circuit that converts the alternating current into a direct current. The [types of rectifiers](#) are as follows;

- **Uncontrolled rectifier;** these rectifiers are made from diodes & their output only depends on the input supply. They are not controlled by any external means
- **Half wave rectifier;** that converts half i.e. positive half or negative half-wave of the AC cycle into DC.
- **Full-wave rectifier:** it converts both positive & negative half of AC into DC.
 - **Bridge rectifier;** It is made of 4 diodes to convert full AC wave into DC
 - **Center tap rectifier;** it utilizes a center tap transformer with only 2 diodes to convert full AC into DC.
- **Controlled Rectifier:** these rectifiers are made from SCR (thyristors) & their output voltage can be controlled by varying the firing angle.
 - **Half controlled:** These rectifiers are made from SCR as well as diodes.
 - **Full controlled:** it is purely made from SCR & provides full control over the voltage through the external triggering pulse.

- **What is a Zener diode?**

It is a special [type of diode](#) that not only conducts current in forward but also in reverse direction once the [Zener voltage](#) is reached. It is used in reverse bias to maintain a constant voltage over a range of currents. It is a safety device used as a voltage regulator or to protect other components from overvoltage.

- **Difference between Analog and Digital Circuit**

Analog circuit can process an analog input signal that has continuously varying voltage.

An analog circuit can convert an analog signal into a digital signal.

Digital circuit can only process a digital signal i.e. a signal that has only two levels (1 or 0 / High or low). They are used in logic-based circuit designing to process complicated functions like in computer & cellphones etc.

- **What is a laser diode?**

The Laser diode converts electrical energy into light energy just like an LED but it creates coherent light. It is a PIN junction diode that creates a beam of light by a combination of holes & electrons in the intrinsic region. They are used in optical communication, pointer, printer, etc.

- **What Is the Difference Between Fuse And Breaker?**

Answer:

The **fuse** is made of a metal wire called fuse link or element that melts when the current exceeds its limit. it works automatically & it is a one-time use device that needs to be replaced.

The **circuit breaker** is an electromechanical switch that opens the circuit during overcurrent or short-circuits. It works automatically as well as manually & it can be used again by resetting the lever.

Related Post: [Main Difference between Fuse and Circuit Breaker](#)

- **What Is Marx Circuit?**

Answer :

Marx circuit or Marx generator is a circuit that charges a number of capacitors in parallel & then discharges them in series to generate high voltage. It is used for creating high voltage using a low voltage source.

- **What Is The Principle Of Motor?**

Answer:

A current-carrying conductor (loop-shaped) placed inside the magnetic field experiences a mechanical force whose direction is given by Fleming's left-hand rule.

- **Which Motor Has High Starting Torque And Starting Current Dc Motor, Induction Motor Or Synchronous Motor?**

The DC series motor has the highest starting torque out of all motors & that is why they are used in electrical machinery requiring high starting torque like cranes, hoist, etc.

- **What Is ACSR Cable And Where We Use It?**

ACSR stands for "Aluminum Conductor Steel Reinforced". Its outer strands are made of highly pure aluminum to have better conductivity while the center strands from steel to increase the tensile strength of the cable. It is used for overhead transmission lines. It has better conductivity & low weight as well as cost.

- **What Is a Vacuum Circuit Breaker?**

Answer:

A circuit breaker breaks the circuit by opening the contact terminals. During the opening, an arc is generated between the terminals that can be quenched using various mediums. In VCB, the medium for arc quenching is a vacuum. The vacuum has a high voltage arc quenching ability as compared to air & they are used for in high voltage circuits.

- **What Is The Difference Between Surge Arrestor And Lightning Arrestor?**

Answer:

Both of them are used as protection devices for grounding the high voltage transients or surges. The surge arrestor is used inside the circuit to protect the components from high voltage spikes. The lightning arrestors are used outside the circuit such as on transmission tower to protect them from high voltage strikes of lightning.

- **Why Use The VCB At High Transmission System? Why Can't Use ACB?**

Answer:

VCB stands for Vacuum circuit breaker & ACB stands for Air Circuit Breaker. The VCB uses the vacuum as its arc quenching medium While ACB uses air as its arc quenching medium. The vacuum has very fast arc quenching & recovering properties as compared to ACB thus they are used in HT while ACB is used in LT.

- **What Is Meant By Insulation Voltage In Cables? Explain It?**

Answer:

The **insulation** voltage or insulation level is the rated voltage of a cable at which the cable can withstand it for its lifespan without rupturing or breakdown.

- **What Are The Advantages Of Star Delta Starter With Induction Motor?**

Answer:

The main advantages of star delta starter are;

- To decrease the starting current required for the induction motor because it is 6 to 7 times higher than full load current which can damage the windings of the motor.
- To eliminate the voltage drop problem because of the huge amount of starting current results in a voltage drop along the consumer line which may damage other electrical appliances.
- Its operation is very simple
- The cost of this starter is comparatively very cheap.
- It has a good torque to current performance.
- **What Is the Difference Between Resistance Grounding System And Resistance Earthing System?**

Answer:

Resistance Grounding; In this system, the load neutral is connected to the ground using resistor/s to limit & carry the **fault current** in case of unbalanced conditions.

Resistance Earthing; this is used to protect the equipment from fault condition by carrying the current through **earth cable**.

What Is Critical Disruptive Voltage?

Answer:

It is the minimum voltage required for the breakdown of the insulation (air) between a phase & neutral to discharge the current. It is the voltage at which the corona discharge starts.

- **How Many Types of Faults Occur in a 3 phase Power System?**

Answer:

The faults in the three-phase system are named below:-

Open circuit fault:

- Single-phase open fault
- Double phase open fault
- Three-phase open fault

Short circuit fault:

- Single line to ground fault (LG)
- Double line to ground fault (LLG)
- Line to line fault (LL)
- Three phase short circuit fault (LLL)

- **What Is Skin Effect?**

Answer:

The current density of the AC passing through a conductor tends to be very high near the surface & very low near the middle of its cross-section area. This phenomenon is called skin effect & directly proportional to the frequency. That is why; stranded conductors are used for power transmission

- **What is latching current & Holding current?**

The minimum initial current required to maintain thyristor in forward conduction state immediately after turning ON is called latching current. The gate signal is removed after providing a latching current.

The holding current is the minimum current required to maintain the thyristor in forward conduction otherwise the thyristor will return to forward blocking state.

The latching current is the current required to switch on the thyristor while the holding current is the minimum limit for turning off the thyristor. The latching current is always greater than holding current.

- **Why is thyristor SCR considered as a Charge controlled device?**

A **thyristor** is triggered into forward conduction state when the charge is injected by applying the gate signal (voltage between gate & cathode terminal). As the SCR is switched on by the current or charge, it is known as a charge controlled device.

- **What is meant by knee point voltage?**

The Knee point voltage of a current transformer is the minimum voltage at which a minimum increase (like 10%) of voltage will increase the magnetizing current by a large margin (like 50%).

- **What is reverse power relay?**

A reverse power protection relay is a directional protection device that protects the generator from taking current back from the grid & causes what is known as a generator motoring effect. it can damage the generator.

- **What will happen if a DC supply is given to the primary of a transformer?**

A transformer is essentially an inductor (coil) that has high inductance & low resistance. For a DC supply, there will be no inductance & there will be no mutual induction between the windings. Also, the low resistance will draw a huge amount of current that will damage the coil & the insulation. Thus, a **transformer won't operate on DC** and may explode with fire.

- **What are the different methods for starting an induction motor?**

The methods used for starting an induction motor are the following:

- DOL: direct online starter
- Star delta starter
- Autotransformer starter
- Resistance starter
- Series reactor starter

- **What are the various kinds of cables used for transmission?**

The **types of cables** based on the transmission voltage ratings are given below ;

- **Low tension** or LT cable is used for transmission of voltage below 1000v
- **High tension** or HT cable used for transmission of up to 11kv.
- **Super tension** cable is used for handling voltage up to 33kv
- **Extra high tension** cable can handle voltage up to 66 kV.
- **Extra super tension** cable is used for transmission of voltage up to 132kV.

- **What is the slip of an induction motor?**

The percentage difference between the synchronous speed N_s & the rotor speed N of an induction motor is called slip. it is denoted by S . The rotor speed of the induction motor is always less than its synchronous speed.

Even More Electronics Engineering Questions and Answers with Explanation

19) Mention the difference between Analogue and Digital circuit?

Analogue	Digital
• These circuits operate on continuous valued signals • No conversion of the input signal required before transmitting, the circuit directly executes various logical operations and produces an analogue output • There is no probability of losing any information as there is no conversion • Analogue lacks flexibility	• These circuits operate on the signal at two level 0's and 1's • Before the signal is transmitted, it is converted into digital form. • During signal conversion, amount of information is lost • Digital circuits anticipate high flexibility

Explain what is laser diodes?

Laser diodes are compact transistor like packages with two or more electrical leads. Lasing occurs when stimulated emission results into the amplification of photon confined to the lasing mode. These photons hit back and forth between the back and front mirror, and hence a diverging beam emits from the laser diode packages.

These interview questions will also help in your viva(orals)

What is meant by reverse polarity and how it can be fixed?

Reverse polarity is referred in a condition where one or more of your receptacles are connected incorrectly. To fix the reverse polarity, check the wire connection at the outlet and inspect your receptacle. A receptacle with reverse polarity will have the white wire screwed to the hot side and the black wire will be connected to the neutral side, if that the case swap the wires and it will resolves the problem. If it persists, a licensed electrician will be needed.

Explain what rectifiers is and what are the types of rectifiers?

A rectifier is an electrical device that transforms A.C or alternating current into direct current (D.C), which flows in only one direction. The types of rectifiers are

- Half wave rectifier: It uses one p-n junction
- Full wave rectifier: It uses two p-n junction

Explain what is Zener diode?

Zener diode is a type of seme-conductor diode that allows current to flow in the opposite direction when exposed to enough voltage.